Leandro Baroca

BTEC Level 3 (QCF) Certificate in IT Subsidiary Diploma in IT 90-credit Diploma in IT Diploma in IT Extended Diploma in IT Faculty of Art, IT & Media Highlands College Jersey Centre No: 67216	
	Year: 2016 / 2017

Unit No(s):	Unit title(s)
Unit 2	Computer Systems

Assessor(s)	Internal Verifier(s):
R Elvidge	S Taylor

Duration (minimum number of hours required to complete assessment evidence between issue date and deadline)
6 hours

Sources of Information (textbooks / websites / lesson resources)
GTS A+ 220-801 & 802 Material (Moodle)

Scenario / Vocational Context

You work in a large computer store as a trainee technical sales person and you advise customers and other members of staff on computer systems. A major part of your job is to give expert advice to customers as to which system to purchase based upon their specific needs. Another part of your job is to train staff members so that they can also advise customers as to the system specification to choose based upon their particular needs.

Instructions

Task 1. [Unit 02 P1]

Complete the tables on the following pages to illustrate your knowledge of the functions of a typical PC. You will need to use actual images of motherboards to identify and locate appropriate components.

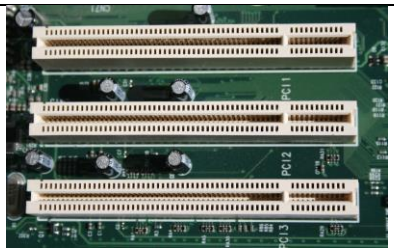
	Image / Reference	Function	Form Factor	Compatible CPU
Motherboard	Asus Maximus VIII Hero Intel Z170 ATX Motherboard 	The motherboard is the foundation of a computer in which it allocates power and allows communication to the CPU, RAM and all the other hardware components.	ATX Form Factor 30.5 cm x 24.4cm (12 inch x 9.6 inch)	Intel Core i7-6700K Skylake Quad Core 4.0GHz (4.2GHz Turbo) CPU
BIOS/UEFI		BIOS is a system which boots up the operating system on the computer. UEFI is an interface which it connect the PC's firmware to its operating system.		

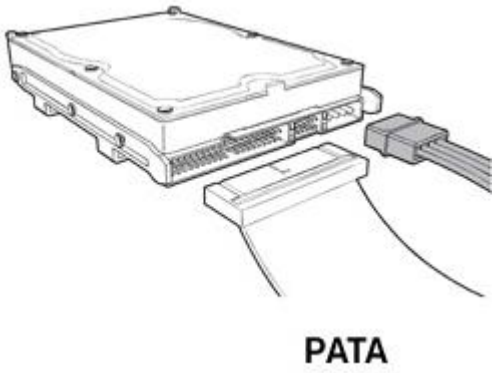
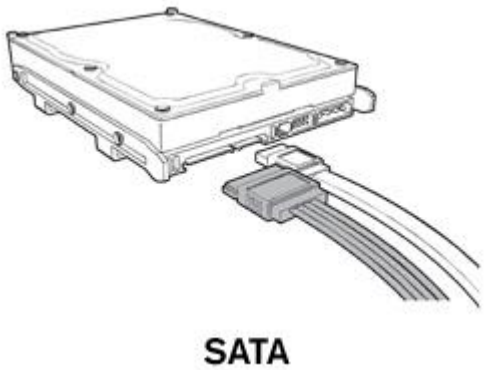
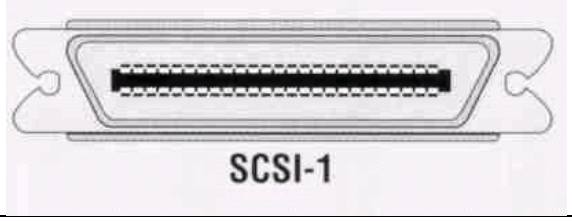
	Image/ Reference	Function	Compatible Motherboard	Socket Type
CPU	Intel Core i7-6700K Skylake Quad Core 4.0GHz (4.2GHz Turbo) CPU 	The CPU carries out instructions of a computer system. I.e. input/output, logical operations, arithmetic operations, etc.	Asus Maximus VIII Hero Intel Z170 ATX Motherboard	FCLGA1151
Power supply		The PCU converts AC power to low-voltage, regulated DC power for the components of the PC.		
Fan and heat sink or cooling for CPU		The HSF help cool the CPU of the PC to prevent the CPU from overheating. Active heatsink is usually a fan type ; passive is an aluminium radiator.	Asus Maximus VIII Hero Intel Z170 ATX Motherboard	

Ports	Image / Reference	Function	Number of Pins	Transfer Rate
Parallel		The function of a parallel port is to connect computers, printers and other devices with demand for high bandwidth.	25 pins	150 Kbps

Ports	Image / Reference	Function	Number of Pins	Transfer Rate
Serial		A serial port sends data by one bit at a time. It was commonly used for a mouse and modems, but it can be used for networks and printers.	9 pins	9600 Bps
USB 2.0		A USB transmits, receives or controls information over the bus. It is possible to implement multiple functions into a USB and embed a hub with a single USB cable.	4 pins	480 mbps / 60 MBps
USB 3.0				5gbps / 640MBps
Thunderbolt		Thunderbolt has many functions. For example, it can be used for: Video Capture & Playback, SCSI, super-fast graphics and other PCI express cards.	20 pins	Thunderbolt 1: 10Gbps Thunderbolt 2: 20Gbps

Memory	Image / Reference	Function	Number of Pins	Memory Speed (Hz)	Maximum Data Rate (Bps)
ROM		ROM is a permanent storage of memory which cannot be modified, unlike RAM.			
SRAM (cache)		Static RAM is RAM that keeps bits in its memory as long as the power is on.			
DDR3 SDRAM		Synchronous DRAM is a type of dynamic RAM that is synchronized with the clock speed of the microprocessor in which the RAM is optimized for.	240 pins	800 MHz	2133 Mb/s
DDR4 SDRAM			288 pins	1600 Mhz	3200 Mb/s

Expansion Slots	Image /Reference	Function	Transfer Rate
PCI		The PCI transmits both address and data signals, sending the address on one clock cycle and data on the next. Burst data can be sent starting with an address on the first cycle and a sequence of data transmission on a certain number of successive cycles.	533 MBs
PCIe		PCIe transfers data over two signal pairs; two wires for transmitting and two wires for receiving. It has lanes which are capable of sending and receiving 8-bit data packets simultaneously between two points.	500 MBs
AGP		The AGP is a high-speed port attaches the video card to a computer system. With the AGP you are able to play more complex programs with high demands.	533 MBs

Interfaces	Image /Reference	Function	Transfer Rate	Number of Devices
PATA		PATA is a standard interface for the connection of storage devices, i.e. optical disk drives, hard disk drives, etc.	133 MBs	2
SATA		SATA is a bus interface that connect mass storage devices such as hard disk drives, optical drives and SSDs to a bus interface.	600 MBs	2
SCSI - 1		SCSi is a parallel interface that is attacked to printers, disk drives, scanners	5 MBps	7 devices + 1 adapter

Interfaces	Image /Reference	Function	Transfer Rate	Number of Devices
		and other peripherals.		
Ultra320 SCSI		Ultra320 SCSI has the same function as the SCSI-1 but with much faster speeds.	320 MBps	15 devices + 1 adapter

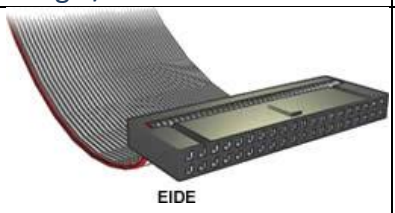
Input Devices	Image /Reference	Function	Connection type
Keyboard		The keyboard helps the user input data to the computer.	USB
Mouse		The mouse helps the user navigate through the software.	USB
Scanner		The scanner captures images from a photographic display into a computer editing display.	USB
WebCam		The webcam is used to film the user, it's uses vary from group calls to filming content to upload online as entertainment.	USB

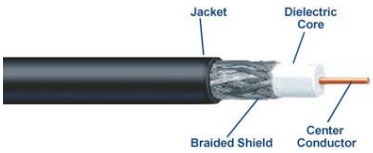
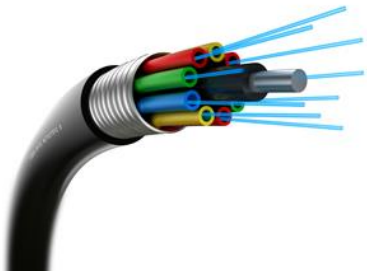
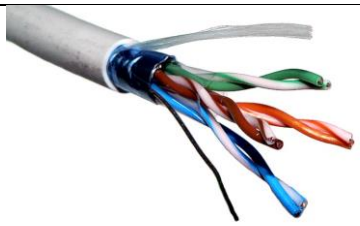
Monitors	Image /Reference	Function	Screen Resolution	Screen Size	Connection Type
CRT		The CRT monitor has a vacuum tube that contains electron guns and a phosphorescent screen which displays images.	640x480 pixels	19 inch	VGA
Flat Screen e.g. LCD		The LCD monitor uses the light-modulating properties of liquid crystals to emit light (not directly) through the pixels.	1920 x 1200	24 inch	HDMI / DVI

Printers	Image /Reference	Function	Pages/Minute	Connection Type
InkJet		The inkjet printer creates a digital image by propelling droplets of ink on to paper, plastic, etc. Inkjet is the most common printer used.	InkJet printers can print up to 6 to 20 pages per minute.	USB
All-in-One		The AiO printer operates the functionality of multiple devices in one. The AiO printer can have based on an inkjet printer or a laser printer.	All-in-One printers can print up from 12 to 45 pages per minute.	USB / Ethernet / Wi-Fi

Backing Storage	Image /Reference	Function	Fixed or Portable	Typical data transfer rate	Typical Capacity
Magnetic Disk		The HDD is used for storing and retrieving digital information using rigid, rapidly rotating disks.	Fixed	30 MBps	2 TB
Solid State Drive		The SSD has the same function as a HDD but, unlike HDDs, SSDs don't have any moving parts.	Fixed	1500 MBps	250 GB
Pen drives		Pen drives can be used for storage, data back-up and transfer of computer files.	Portable	120 MBs	64 GB
Flash memory cards / SD Card		Memory cards stores data such as text, pictures, audio and video.	Portable	30 MBs	32 GB
Optical Media					
CD		Discs are capable of storing large amounts of data on one disc the size of a compact disk.	Disc: Portable ROM: Fixed	1.2 MBs	700 MB
DVD			Disc: Portable ROM: Fixed	1.321 MBs	4.7 GB

Backing Storage	Image /Reference	Function	Fixed or Portable	Typical data transfer rate	Typical Capacity
Blu-Ray			Disc: Portable ROM: Fixed	54Mbps	50 GB

Cabling	Image /Reference	Function	Image of connector	Pins/Conductors
EIDE (PATA) Data Cable		It connects devices together like the hard driver and the motherboard to share data via parallel connection.		40
EIDE Power Cable and connector (Molex)		The function of this cable is to power the PATA hard drive.		4
SATA data cable		Serial ATA connects the host bus adapters to mass storage devices such as HDD, optical drives and SSDs.		7
SATA power cable		The function of this cable is to power the SATA hard drive.		15
Network Cables				

Cabling	Image /Reference	Function	Image of connector	Pins/Conductors
Co-axial		Coaxial cable can be used for Ethernet and other types of local area network.		1
Optical		Optical fibre cables are used to transfer signals using light through thin, flexible glass fibres. This method of data transmission is extremely fast because light is the fastest traveling thing in the known universe.		1
Twisted Pair		STP cables are used in older telephone networks, as well as network and data communications.		8

Criteria

To achieve the criteria, you must show that you are able to:

Unit	No	Criteria	Evidence	Student Tracking
2	1	explain the function of computer hardware components	Complete the following tables: Motherboard CPU Ports Memory Expansion Slots Interfaces Input Devices Monitors Printers Backing Storage Optical Media Cabling	